

Multi-gate Mixture-of-Experts (MMoE)

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研究动机

- A real recommendation system often needs to **optimize multiple objectives at the same time**. For example, when recommending movies for users to watch, we may want the users to not only **purchase and watch the movies**, but also to **like the movies afterwards** so that they'll come back for more movies.
- The prediction quality of commonly used multi-task models is often **sensitive** to the data distribution differences and the relationships among tasks. The inherent **conflicts** from task differences can actually **harm** the predictions of at least some of the tasks, particularly when model parameters are extensively shared among all tasks.
- It is therefore important to study the modeling **tradeoffs** between task-specific objectives and inter-task relationships.

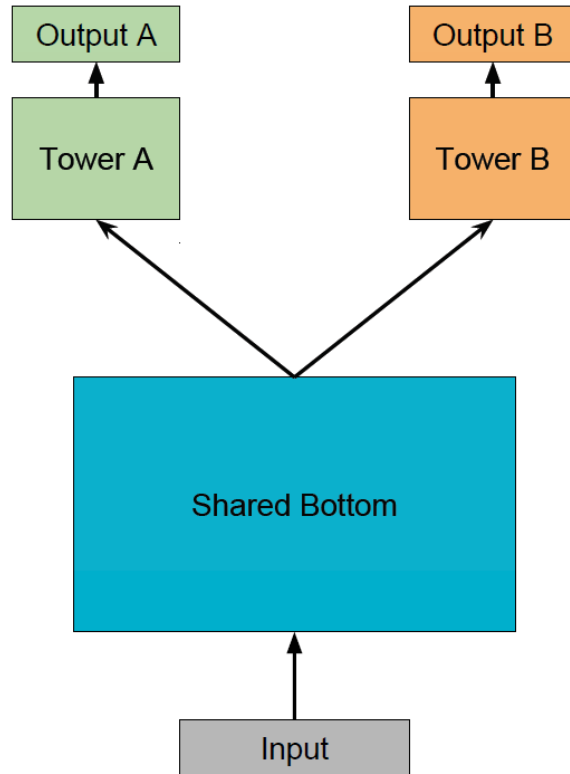
介绍

- 研究问题：多任务学习（multi-task learning）
Build a single model that **learns multiple goals and tasks simultaneously**
- 模型名称：Multi-gate Mixture-of-Experts (MMoE)

相关工作

- Robert A Jacobs, Michael I Jordan, Steven J Nowlan, Geoffrey E Hinton. **Adaptive mixtures of local experts**. Neural Computation 1991.
 - Mixture-of-Experts (MoE)
 - One-gate Mixture-of-Experts (OMoE)
- Rich Caruana. **Multitask learning**. In *Learning to learn*. Springer, 95-133. 1998.
 - Shared-Bottom multi-task DNN structure

Shared-bottom Multi-task Model

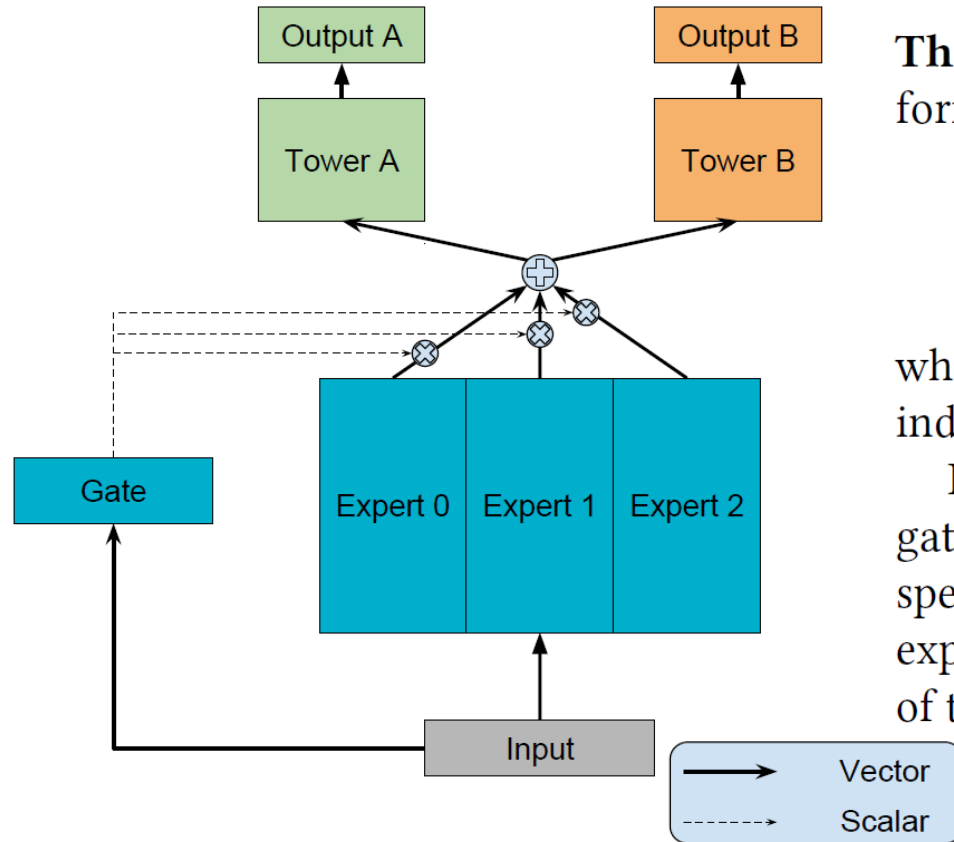


Given K tasks, the model consists of a shared-bottom network, represented as function f , and K tower networks h^k , where $k = 1, 2, \dots, K$ for each task respectively. The shared-bottom network follows the input layer, and the tower networks are built upon the output of the shared-bottom. Then individual output y_k for each task follows the corresponding task-specific tower. For task k , the model can be formulated as,

$$y_k = h^k(f(x)). \quad (1)$$

- This structure substantially **reduces the risk of overfitting**, but can **suffer from optimization conflicts** caused by task differences, because all tasks need to use the same set of parameters on shared-bottom layers.
- It is also called hard parameter sharing.

One-gate Mixture-of-Experts (OMoE) Multi-task Model



The Original Mixture-of-Experts (MoE) Model [21] can be formulated as:

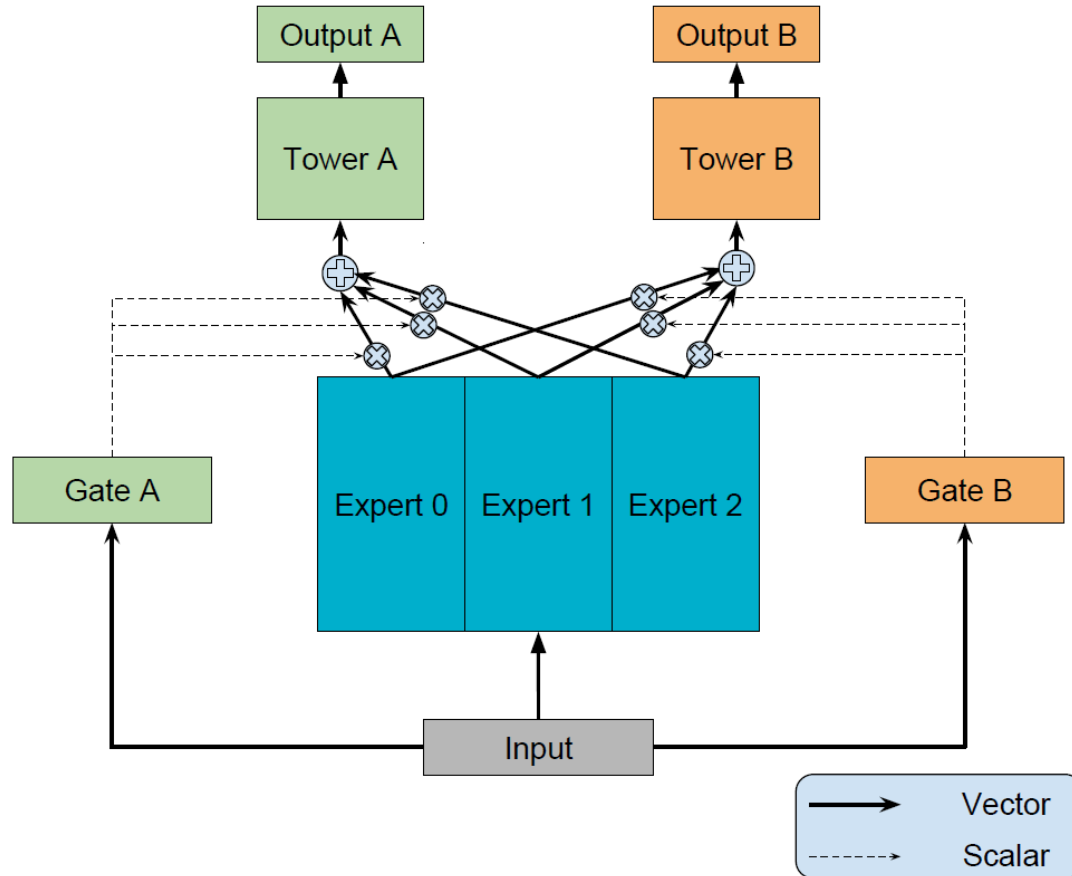
$$y = \sum_{i=1}^n g(x)_i f_i(x), \quad (5)$$

where $\sum_{i=1}^n g(x)_i = 1$ and $g(x)_i$, the i th logit of the output of $g(x)$, indicates the probability for expert f_i .

Here, $f_i, i = 1, \dots, n$ are n expert networks and g represents a gating network that ensembles the results from all experts. More specifically, the gating network g produces a distribution over the n experts based on the input, and the final output is a weighted sum of the outputs of all experts.

- Different tower networks **has the same input**, which may not be the best.

Multi-gate Mixture-of-Experts (MMoE) Multi-task Model



the MoE layer in Eq 5. More importantly, we add a separate gating network g^k for each task k . More precisely, the output of task k is

$$y_k = h^k(f^k(x)), \quad (6)$$

$$\text{where } f^k(x) = \sum_{i=1}^n \underline{g^k(x)_i} f_i(x). \quad (7)$$

See Figure 1 (c) for an illustration of the model structure.

Our implementation consists of identical multilayer perceptrons with ReLU activations. The gating networks are simply linear transformations of the input with a softmax layer:

$$\underline{g^k(x)} = \text{softmax}(W_{gk}x), \quad (8)$$

where $W_{gk} \in \mathbb{R}^{n \times d}$ is a trainable matrix. n is the number of experts and d is the feature dimension.

- Different tower networks **has different input**, since there a specific gating network for each task.

总结

- 主要思路： parameter modulation + ensemble of subnets (experts) + knowledge transfer
Mixture-of-Experts (MoE) + a gating network for each task
- 主要优点：
 - Explicitly models the task relationships
 - Each gating network can learn to “select” a subset of experts
 - Learns task-specific functionalities
 - Avoids the need of adding many new parameters per task as that in previous works
 - The parameter modulation and gating mechanisms can improve the trainability in training non-convex deep neural networks
 - The number of model parameters in the gating networks is negligible

参考资料

- Jiaqi Ma, Zhe Zhao, Xinyang Yi, Jilin Chen, Lichan Hong, Ed H. Chi. **Modeling Task Relationships in Multi-task Learning with Multi-gate Mixture-of-Experts**. KDD 2018.